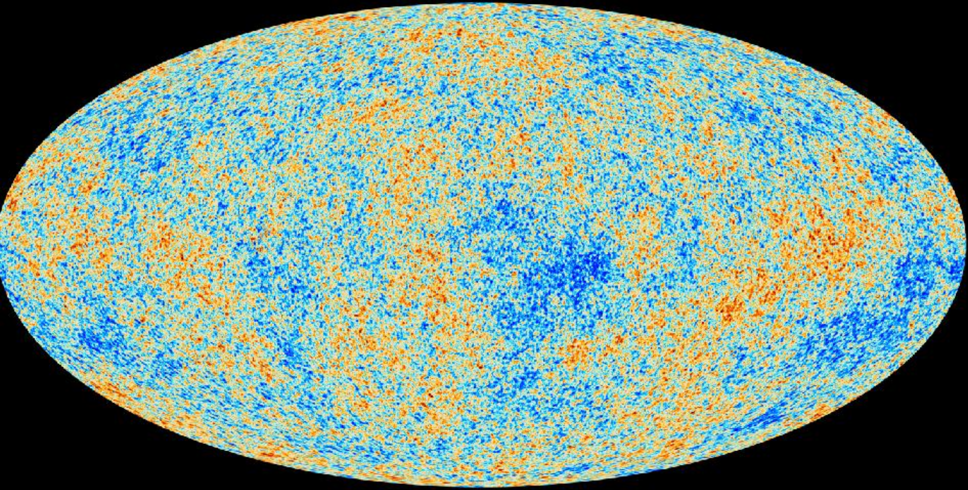

The Big Bang and Inflation

The first 0.00000000000000000000000000000001 seconds

Will Handley
wh260@cam.ac.uk

Astrophysics Department
Cavendish Laboratory
University of Cambridge

20th March 2013

Overview

What is Cosmology?

Cosmology is the study of the universe as a whole

Age: How old is the universe?

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Content:	What are Dark Matter and Dark Energy?

The Night Sky

The Milky Way

The Night Sky

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Ground based telescopes

The Coma Cluster

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Space Telescopes

Hubble Extreme Deep Field

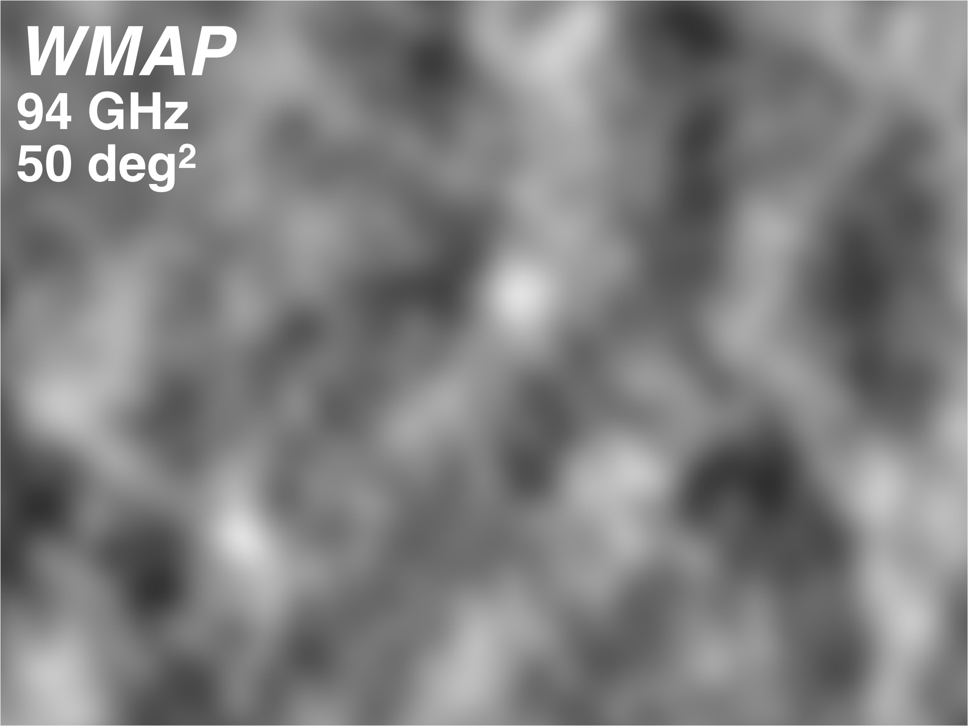
Space Telescopes

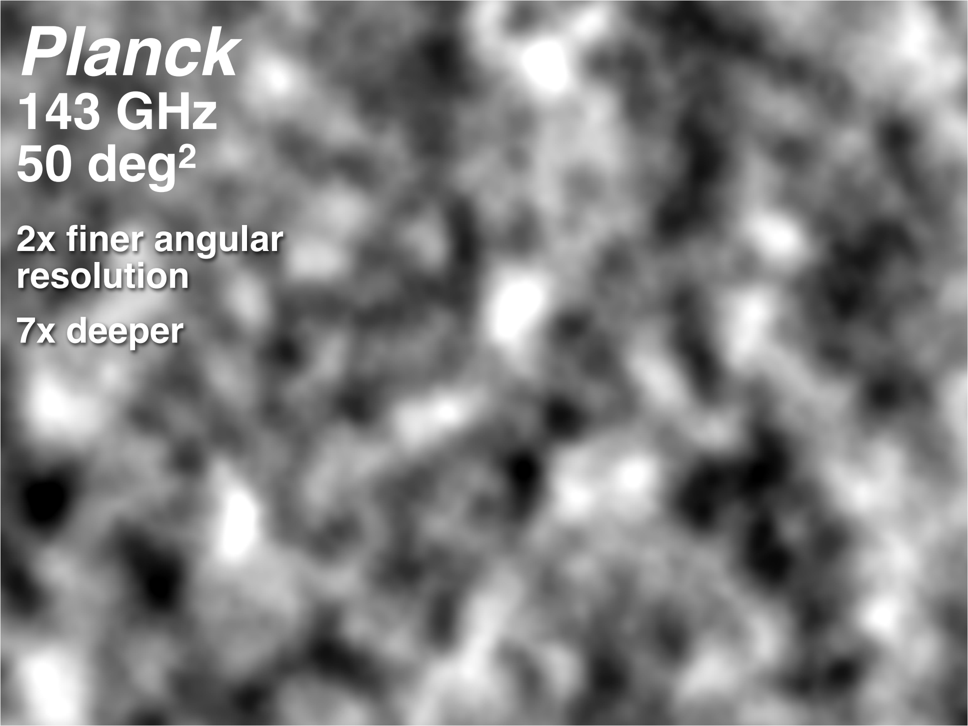
Hubble Extreme Deep Field

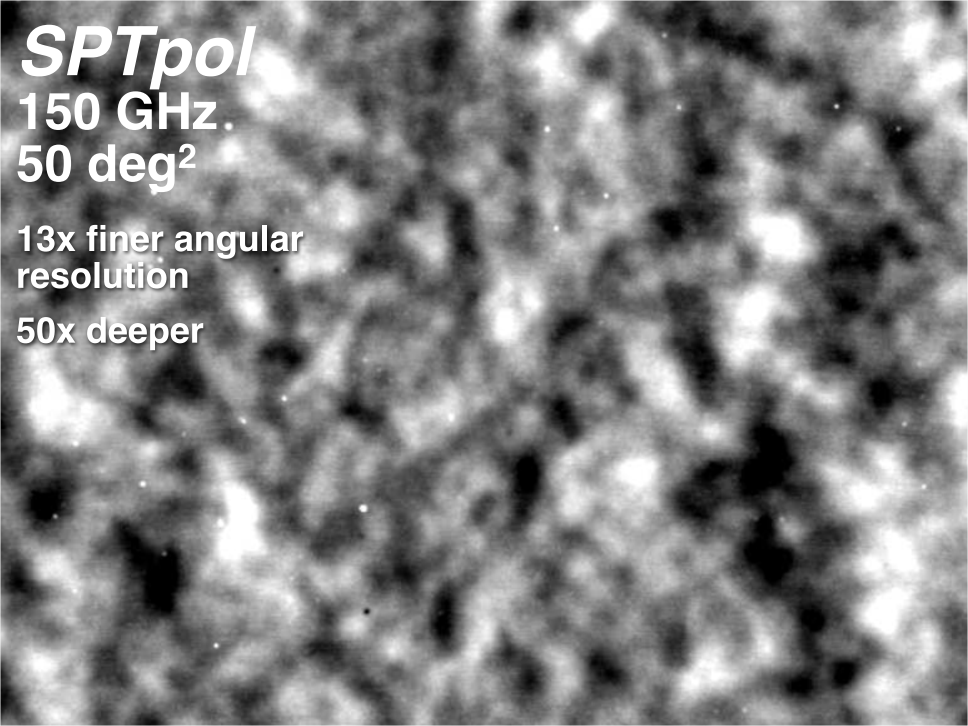
The Dark Ages

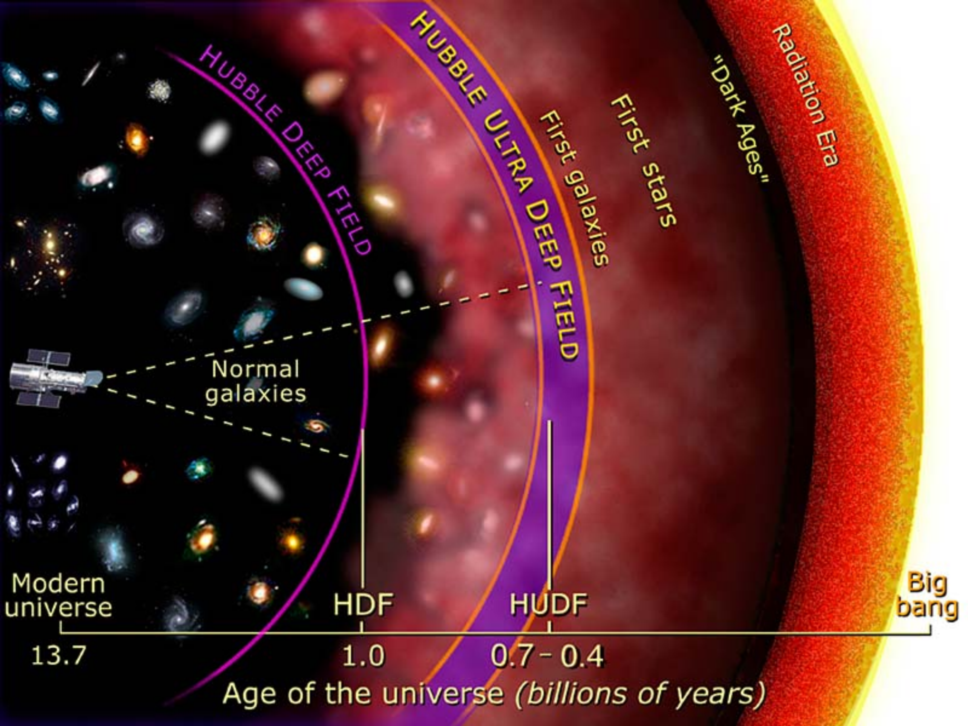
The CMB

Cosmic Microwave Background


WMAP
94 GHz
50 deg²


Planck
143 GHz
50 deg²
2x finer angular resolution
7x deeper


SPTpol
150 GHz
50 deg²
13x finer angular resolution
50x deeper



HUBBLE DEEP FIELD
HUBBLE ULTRA DEEP FIELD
First galaxies
First stars
"Dark Ages"
Radiation Era
Normal galaxies
Modern universe
HDF
HUDF
Big bang
13.7
1.0
0.7 – 0.4
Age of the universe (billions of years)

What is the CMB?

- In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with "The Big Bang".

The Big Bang

Galactic Redshift

The Big Bang

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As observed by Edwin Hubble in 1929, it appears that:

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2. The Further away they are, the faster they are moving away

The Big Bang

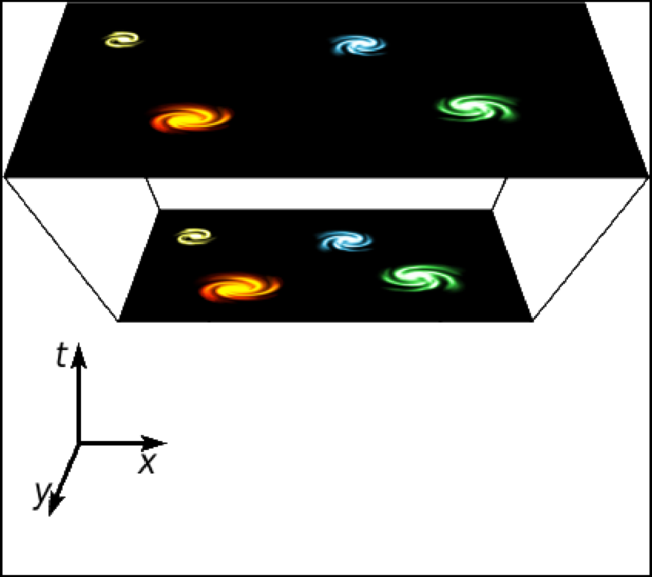
Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

The Big Bang

A stretching universe:



The Big Bang

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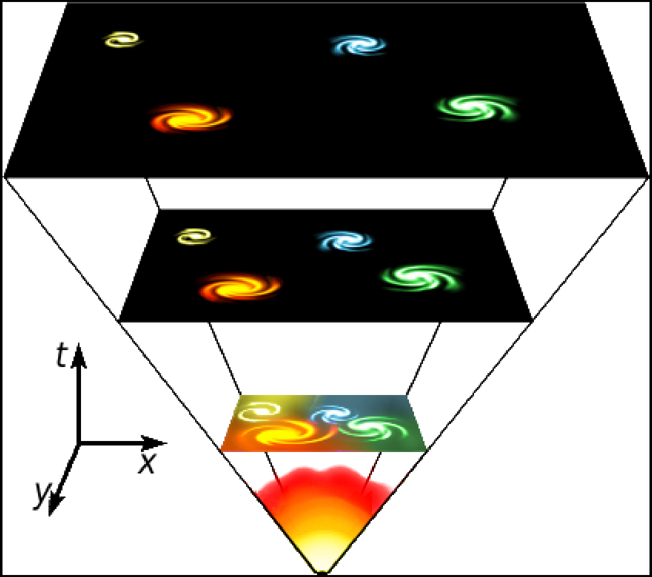

The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:

The Big Bang

Common Misconceptions

- Despite the name, the Big Bang is not an explosion.

The Big Bang

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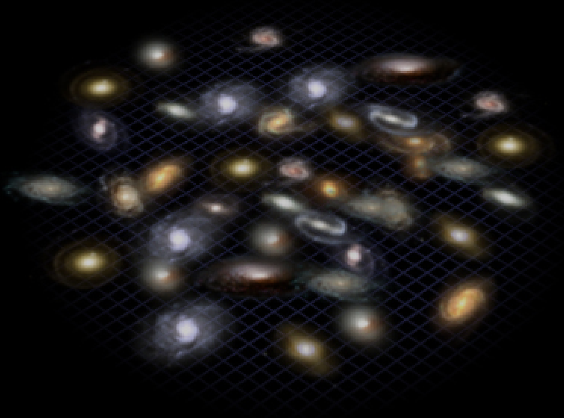
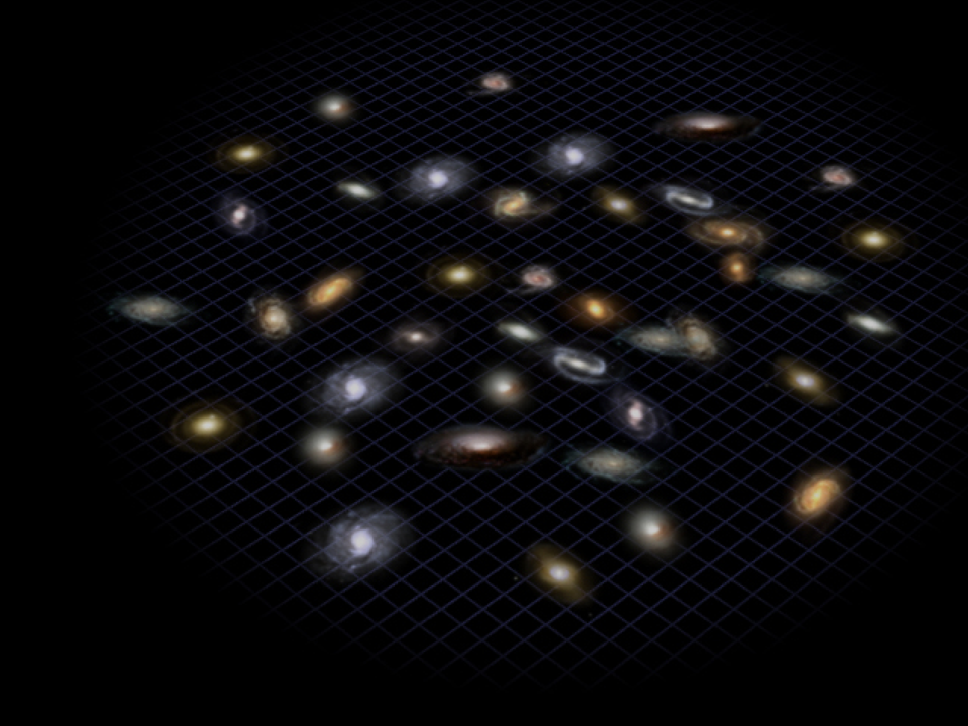
- Despite the name, the Big Bang is not an explosion.
- The Universe is not necessarily expanding into anything.

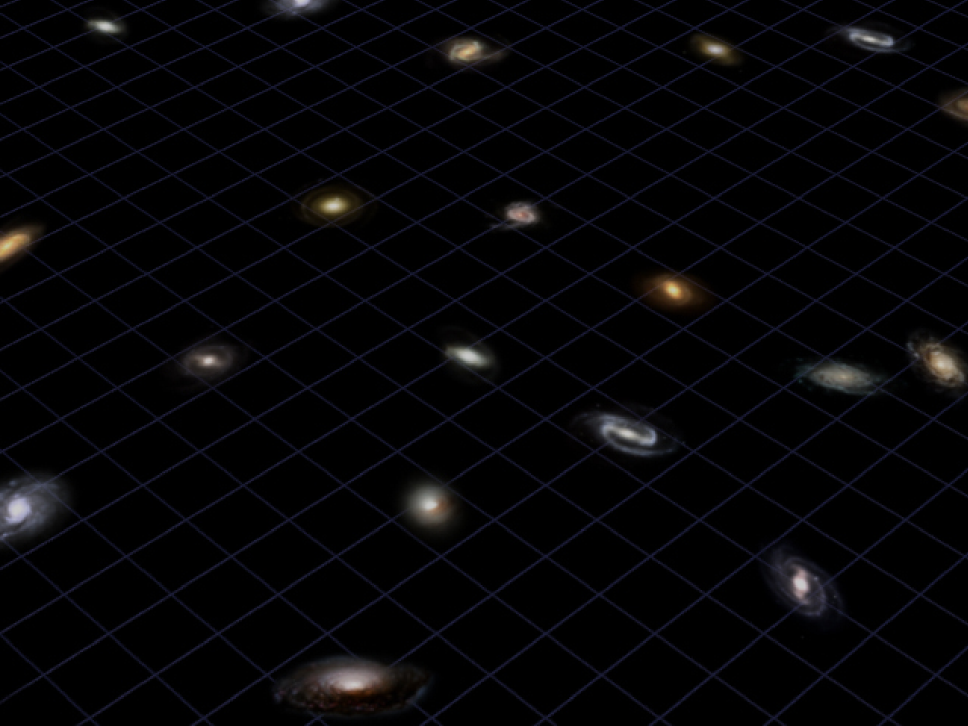
The Big Bang

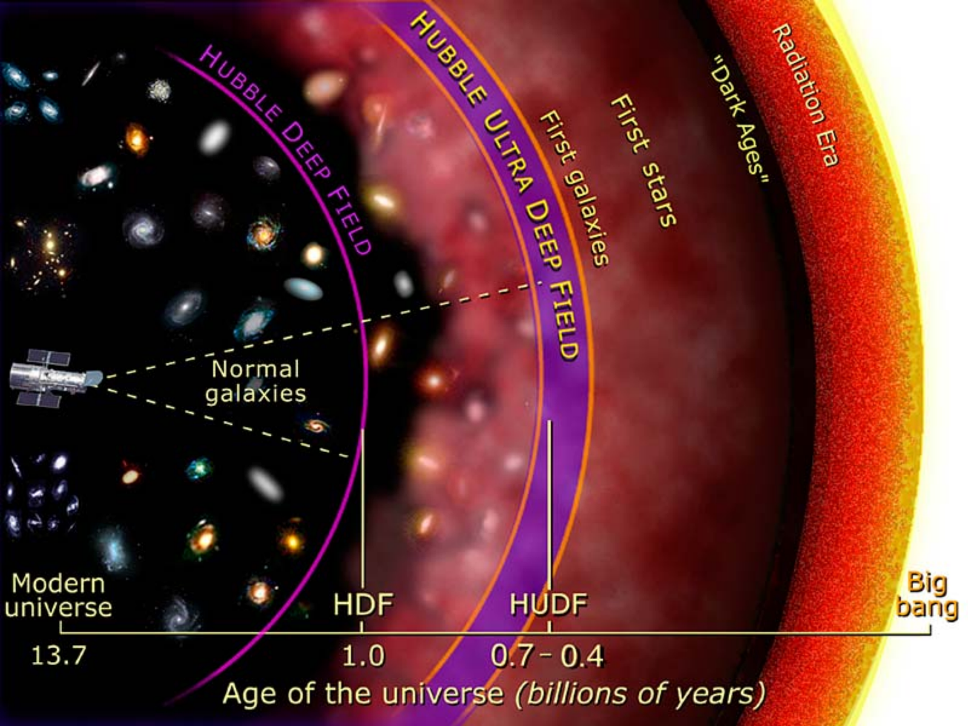
Common Misconceptions

- Despite the name, the Big Bang is not an explosion.
- The Universe is not necessarily expanding into anything.
- The big bang didn't begin at a particular point









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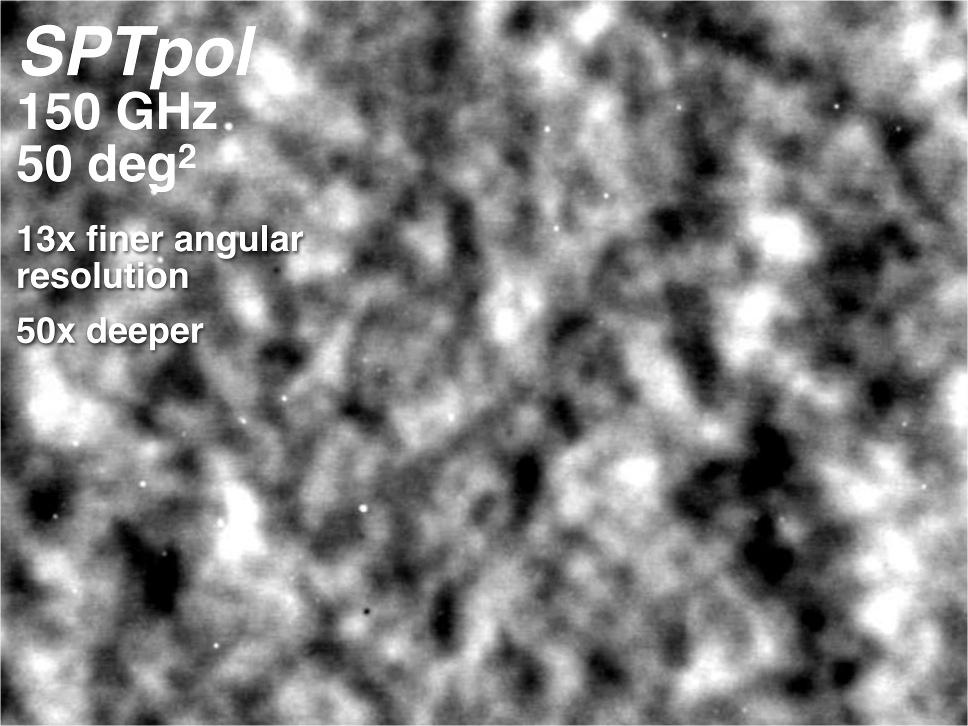
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Current Issues in Cosmology

There are several outstanding problems in Astrophysics which we do not fully understand:

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- Dark Matter

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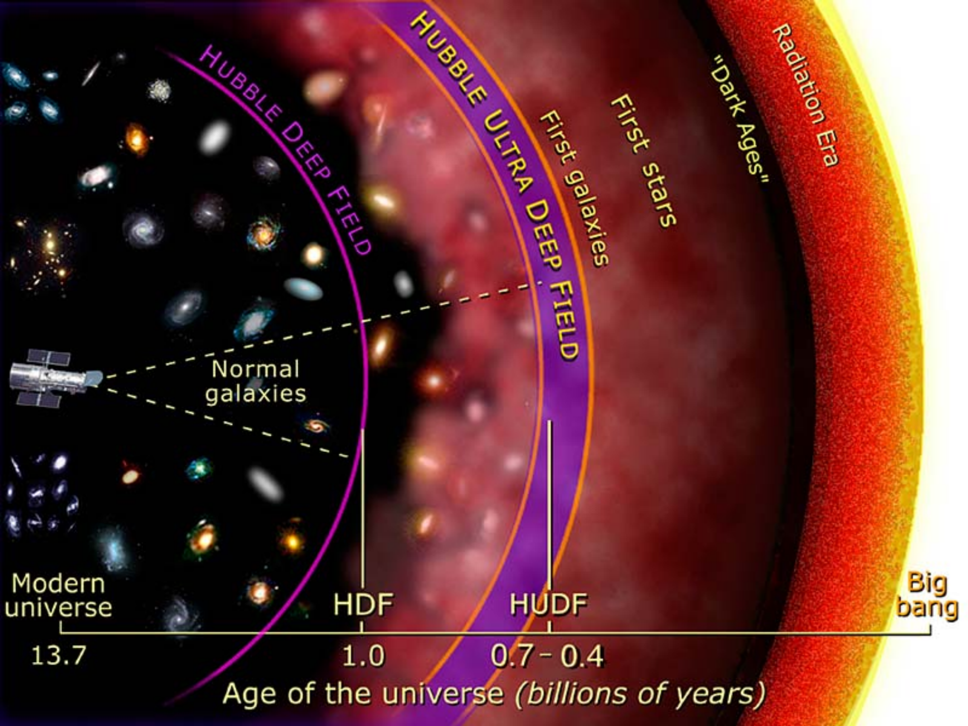
Current Issues in Cosmology

There are several outstanding problems in Astrophysics which we do not fully understand:

- Dark Matter
- Dark Energy
- $t = 0$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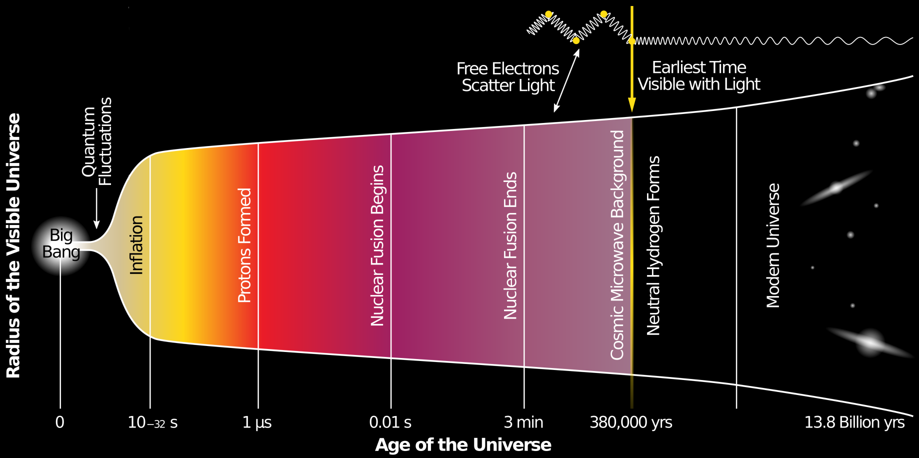
The beginning of time

$t = 0$



Problems with basic cosmologies

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1. The universe is too smooth

Problems with basic cosmologies

1. The universe is too smooth (homogeneous)

Problems with basic cosmologies

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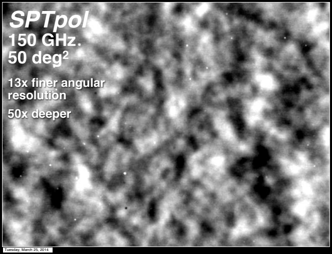
2. The universe is too flat

The universe is too smooth (homogeneous)

- The CMB is far too uniform

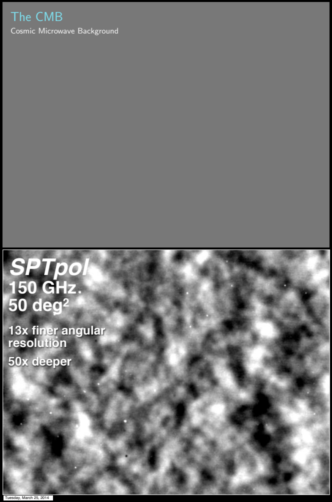
The universe is too smooth (homogeneous)

- The CMB is far too uniform



The universe is too smooth (homogeneous)

- The CMB is far too uniform
- In order for it to be that smooth, the universe would have to be much older.



The universe is too flat

- The universe can come in three shapes

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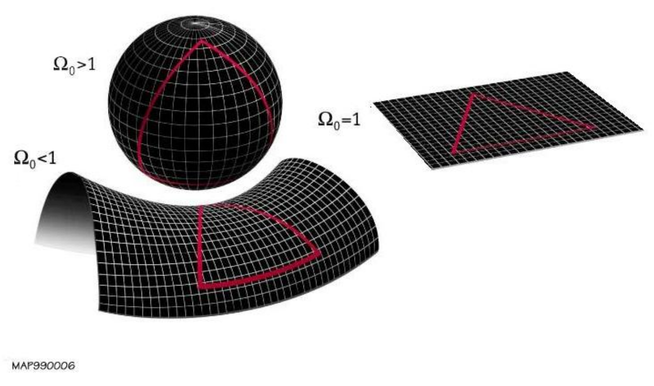

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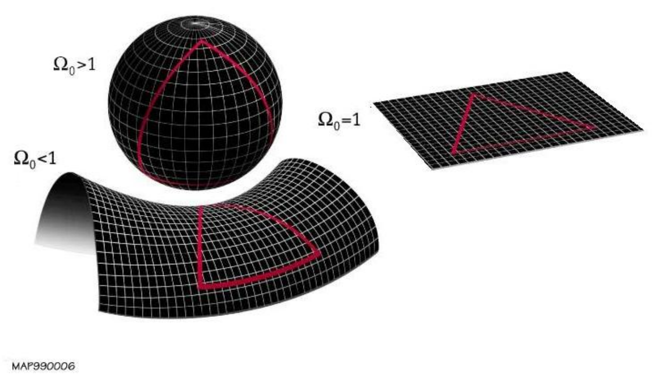

1. Spherical
   (closed)

The universe is too flat

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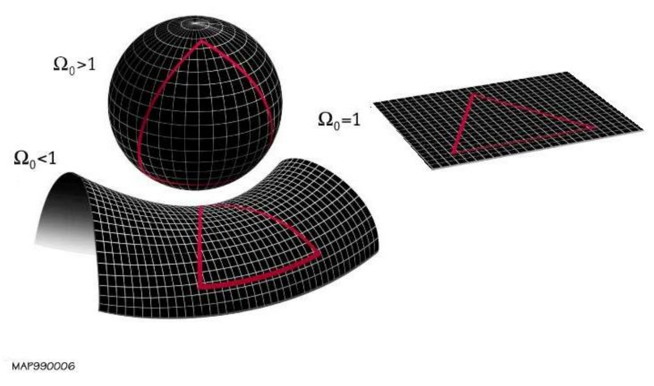

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2. Flat

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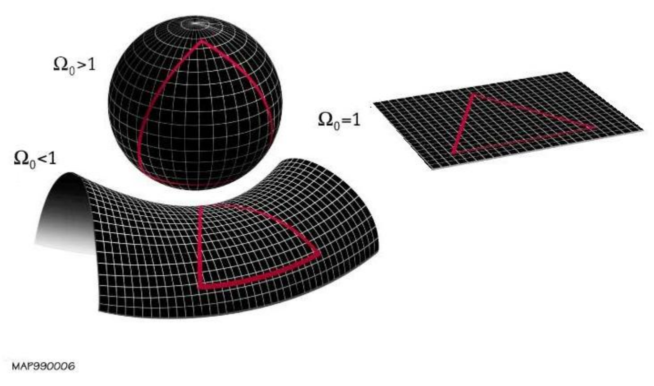

1. Spherical (closed)
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3. Saddle (open)

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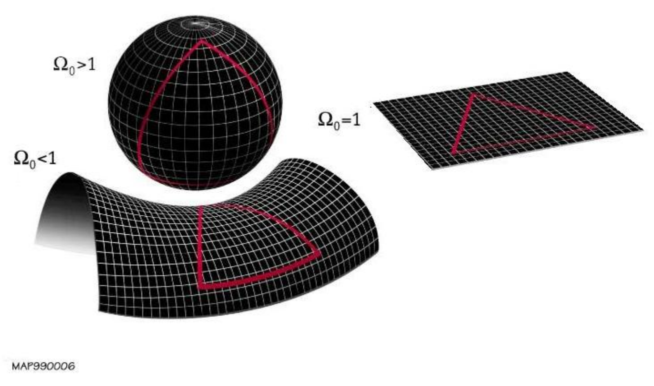

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- Flat is exceedingly unlikely

Inflation

A solution

- Early on in history, the universe expanded incredibly rapidly

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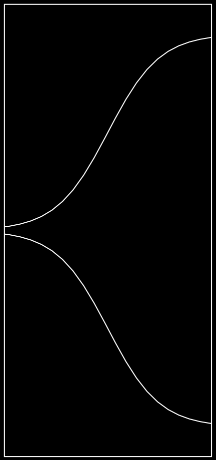

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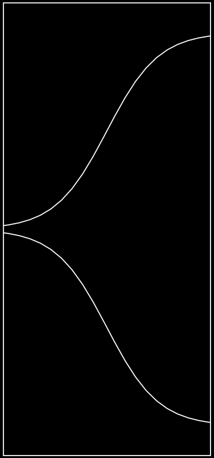
- Early on in history, the universe expanded incredibly rapidly
- We call this accelerated expansion "Inflation"



Inflation

A solution

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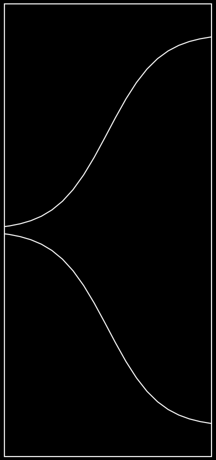
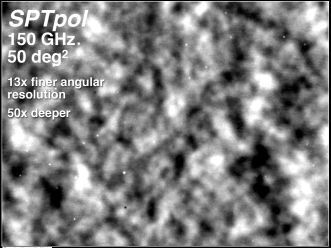
- We call this accelerated expansion "Inflation"
- Virus $\longrightarrow$ Galaxy in 10^{-32} seconds



Evidence for inflation

The CMB

Cosmic Microwave Background



Tuesday, March 25, 2014

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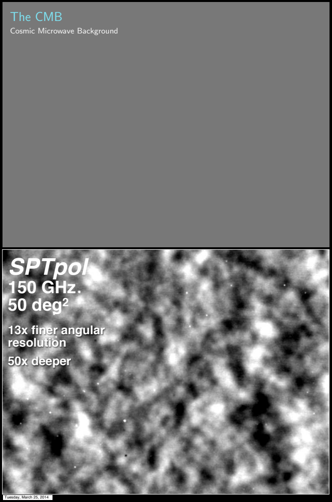

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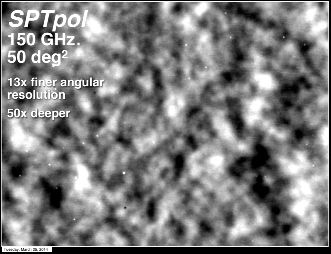
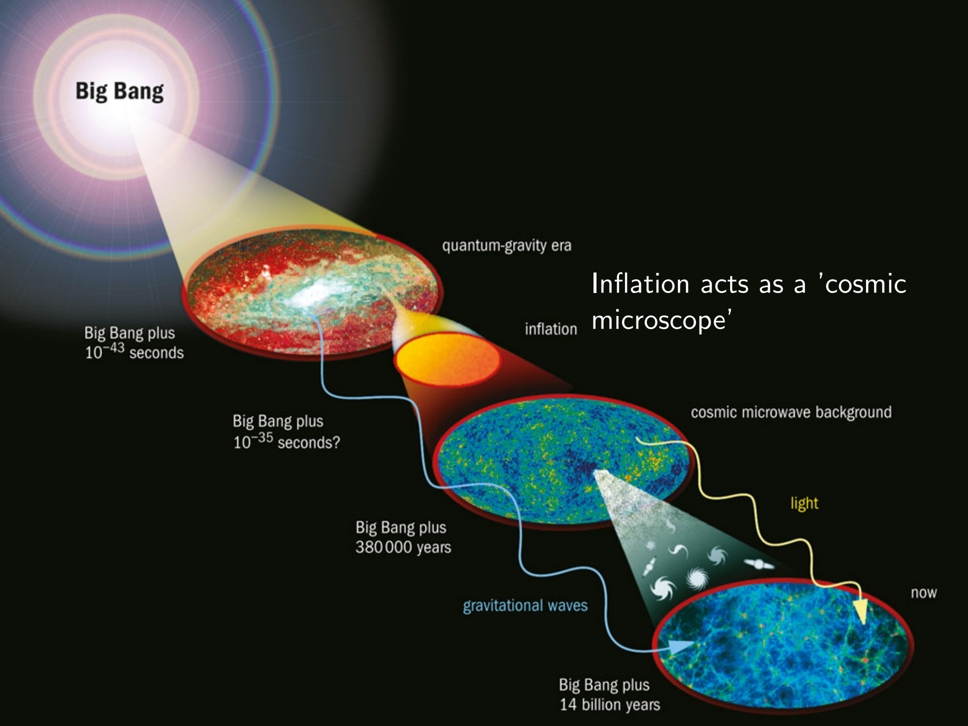
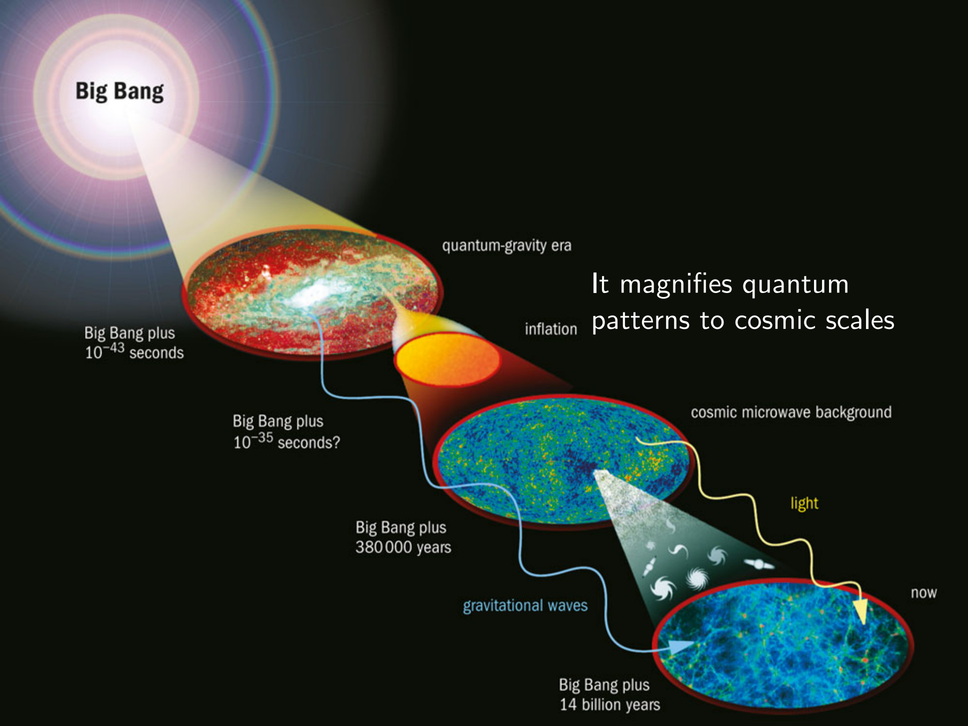


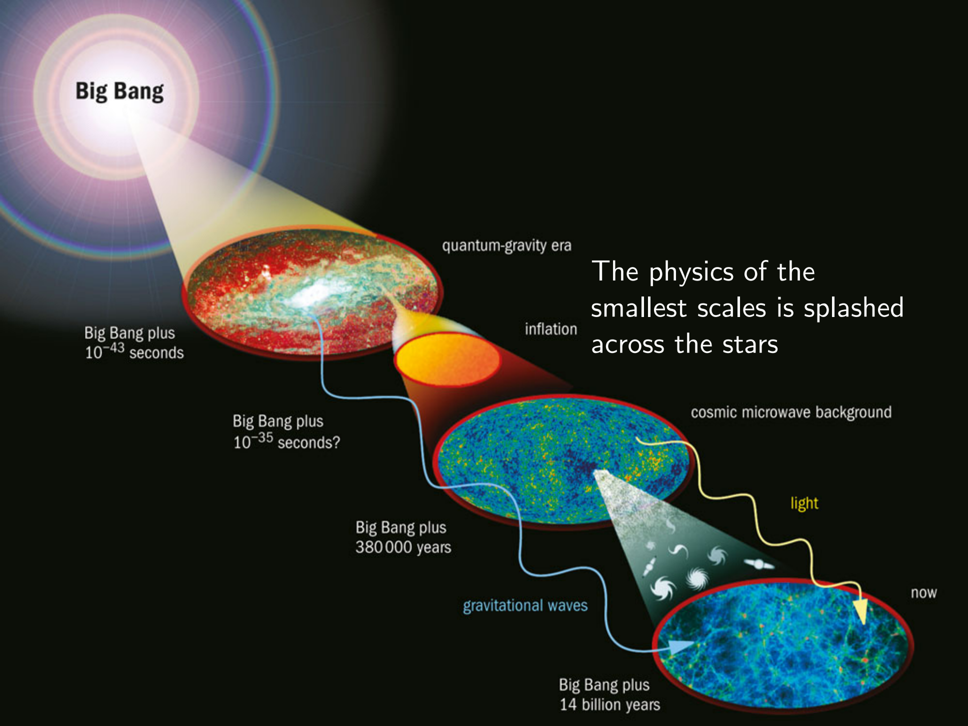
- Not only does inflation explain why the CMB is so smooth. . .
- . . . It also explains the small fluctuations.

Inflation acts as a 'cosmic microscope'



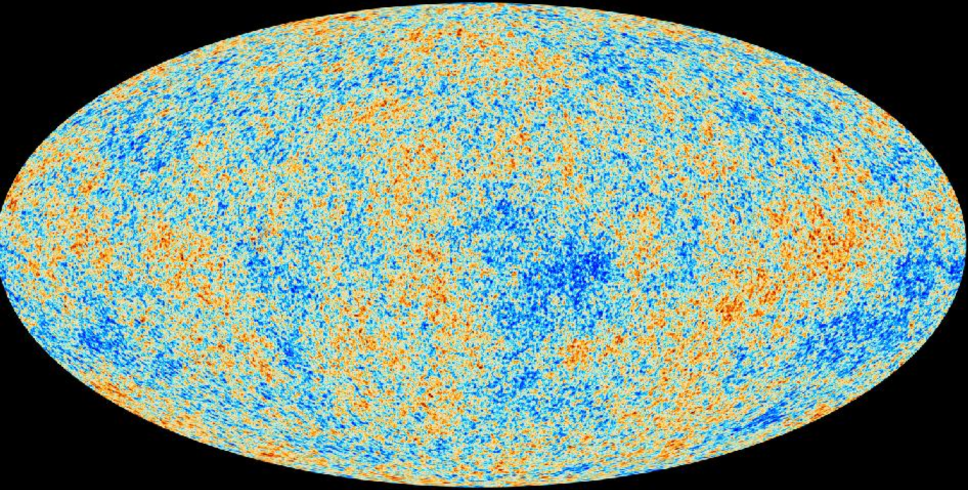

Big Bang
quantum-gravity era
It magnifies quantum patterns to cosmic scales
inflation
Big Bang plus
10^−43 seconds
Big Bang plus
10^−35 seconds?
cosmic microwave background
light
Big Bang plus
380 000 years
gravitational waves
now
Big Bang plus
14 billion years

The physics of the smallest scales is splashed across the stars



Thank you!

Questions?



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